

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



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Comparative Evaluation of Wavelet-Based ECG Signal Denoising Techniques

M.Sc. Thesis in Electrical Engineering — Bioelectric

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***Dedicated to
my loving father and mother,
to whose unsparing efforts I owe
my existence***

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Abstract

Electrocardiogram (ECG) signals are almost always contaminated by noise during acquisition, and although numerous wavelet-based methods have been proposed for ECG denoising, previous studies have usually examined each method under different experimental conditions (database, noise type and intensity, evaluation metrics), which makes a direct comparison of their results difficult.

The aim of this thesis is to provide a controlled comparative evaluation of four denoising methods: plain wavelet thresholding (DWT), empirical mode decomposition combined with wavelet thresholding (EMD-DWT), variational mode decomposition combined with wavelet thresholding (VMD-DWT), and non-local means (NLM) as a reference method. The four methods were compared on nineteen real signals under fully identical conditions -- the MIT-BIH Arrhythmia Database, additive Gaussian noise at two variance levels, and a common set of evaluation metrics. The complete experiment was implemented and validated in MATLAB.

The results show that VMD-DWT significantly and consistently outperforms EMD-DWT (a mean improvement of 1.33 dB in signal-to-noise ratio, with very high statistical significance), and closely approaches the performance of the simpler NLM method. Both decomposition front ends (EMD and VMD) were also found to provide a significant improvement over plain wavelet thresholding, although the improvement is considerably larger for VMD. The best configuration found in this study was VMD-DWT with the Symlet-4 wavelet at decomposition level 3. A quantitative evaluation of QRS-complex amplitude preservation further showed that this ranking was not achieved at the expense of losing diagnostically important morphological detail.

The findings of this research provide practical, evidence-based guidance for selecting an appropriate ECG denoising method under Gaussian noise conditions.

Keywords: 1- electrocardiogram signal processing, 2- wavelet transform, 3- empirical mode decomposition, 4- variational mode decomposition, 5- non-local means, 6- adaptive noise removal.

Chapter 1

Introduction

1.1 Background and Motivation

Cardiovascular diseases are currently among the leading causes of death worldwide, and their timely and accurate diagnosis plays an irreplaceable role in improving patients' quality of life [1]. In this context, the electrocardiogram (ECG), as a non-invasive and widely used modality, provides valuable information about the electrical activity of the heart and helps physicians identify various arrhythmias and other cardiac disorders [2]. A correct ECG recording encodes the timing and amplitude of atrial and ventricular depolarization and repolarization in the familiar P, QRS and T deflections, and since clinical inference rests mainly on waveform morphology rather than on raw voltage alone, the fidelity of the acquisition system to this morphology is not a secondary engineering concern but a decisive factor in patient outcome.

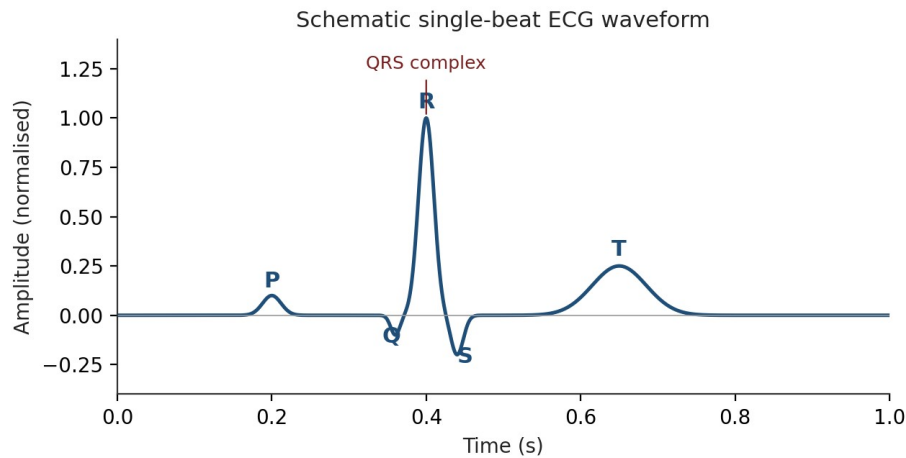


Figure 1.1: Schematic single-beat ECG waveform showing the P, QRS and T deflections referred to throughout this thesis (original illustration; morphological description based on [1], [2]).

In practice, the ECG is rarely recorded under ideal conditions. Power-line interference at 50 or 60 Hz, baseline wander caused by respiration and changes in electrode impedance, electromyographic noise from the surrounding skeletal muscles, and motion artifacts caused by movement of the patient or the electrodes are all superimposed on the cardiac signal during acquisition and transmission [3], [4]. Each of these disturbances occupies a different, and sometimes overlapping, part of the frequency spectrum, which is what makes their

removal difficult: a filter aggressive enough to suppress low-frequency baseline wander equally attenuates the low-frequency components of the T wave, and a filter tuned to remove high-frequency muscle noise risks rounding off the sharp tip of the QRS complex -- on which clinicians rely to measure intervals [5].

Classical denoising methods based on fixed-bandwidth digital filters, although computationally inexpensive, treat the signal as if its statistical properties remained constant over time. The ECG is not stationary in this sense; its spectral content changes from one beat to the next and even within a single beat, so a filter with a fixed, uniform frequency response is structurally mismatched to the problem [5]. The wavelet transform entered biomedical signal processing precisely to resolve this mismatch. Because a wavelet basis is localized in both time and scale, it can represent transient, non-stationary structures -- exactly the kind of structure a QRS complex has -- far more compactly than a Fourier basis, and it is this compactness that allows wavelet thresholding to separate signal-like coefficients from noise-like coefficients without destroying the transient itself [6]. The foundational work of Donoho and Johnstone provided the statistical basis for this separation, showing that a simple thresholding rule applied to wavelet coefficients can achieve a near-optimal trade-off between noise suppression and signal fidelity across a whole class of unknown smooth functions [7], [8], [9].

Building on this foundation, a large body of research has proposed wavelet-based denoising schemes specifically for the ECG. Some of these works refine the thresholding rule itself, such as subband-dependent or data-adaptive thresholds [10]; others optimize the choice of mother wavelet and decomposition depth for cardiac morphology [11]; and a growing branch combines the wavelet transform with a signal-adaptive decomposition front end, notably empirical mode decomposition (EMD) [12] and its more recent variational counterpart, variational mode decomposition (VMD) [13], so that thresholding is applied not to the wavelet coefficients of the raw signal but to those of the intrinsic, approximately mono-component modes extracted from it [14], [15]. At the same time, entirely different denoising paradigms have also been proposed, among them the non-local means (NLM) filter -- originally developed for natural images and later adapted to one-dimensional biosignals -- which estimates each sample as a weighted average of signal patches with similar structure, without any explicit frequency-domain reasoning [16].

The proliferation of these methods is in itself a sign of a healthy research field. Kamal et al. [17] report a correlation coefficient above 0.95 and a signal-to-noise ratio improvement of

more than 17 dB; Yang and Wei [18] report a mean squared error as small as 0.0146 while explicitly preserving the QRS amplitude; Chatterjee et al. [19] report a signal-to-noise ratio improvement of more than 27 dB for a sparsity-based wavelet autoencoder; and Ge et al. [4] report a 39 to 55 percent improvement over earlier baseline methods. Each of these results is, on its own, a genuine contribution. Taken together, however, they raise a question that none of these papers can answer individually: because each study uses its own database subset, its own noise type and intensity and, in several cases, its own definition of the evaluation metric, the reader cannot rank these methods fairly against one another. It is precisely this gap -- not a lack of good methods, but the lack of a level playing field on which to compare them -- that is the main motivation for this research.

1.2 Problem Statement

The main practical difficulty facing a researcher or biomedical engineer who must choose an ECG denoising method for a specific application is that the published literature does not, by itself, support that choice. A method described as superior in one paper is evaluated in another paper on a different subset of the MIT-BIH Arrhythmia Database, under a different noise type and intensity, and using a different subset of quantitative metrics [2], [3]. Comparative claims such as "method A outperforms method B" are therefore almost always claims about performance under two different experimental protocols rather than one shared protocol, and the two are not equivalent.

The problem is compounded by three further considerations. First, hybrid schemes that combine a signal-adaptive decomposition -- EMD or VMD -- with wavelet thresholding are structurally different from single-stage DWT thresholding, yet the two families are rarely evaluated side by side under identical conditions; Lahmiri's comparative study [14] of wavelet thresholding in the EMD domain and in the VMD domain remains one of the few works that address this directly, and even that study does not extend the comparison to a non-wavelet reference method. Second, similarity-based methods such as non-local means are almost never evaluated alongside the same hybrid schemes, so it is unclear whether the added algorithmic complexity of EMD-DWT or VMD-DWT is actually justified by a performance gain over a simpler patch-based filter [16]. Third, the sensitivity of hybrid wavelet methods to internal design choices -- mother-wavelet type and decomposition level -- is rarely separated from the choice of decomposition front end, so when a hybrid method performs

poorly it is unclear whether the front end, the wavelet configuration, or the interaction between the two is responsible.

Accordingly, this thesis addresses the following problem: given four ECG denoising strategies -- plain DWT thresholding, a hybrid EMD-DWT method, a hybrid VMD-DWT method, and the NLM reference method -- all evaluated on the same recordings from the MIT-BIH Arrhythmia Database, contaminated with additive Gaussian noise of identical intensities, and scored with a common set of quantitative metrics, which method provides the best noise suppression without degrading the clinically important morphology of the signal, and which internal design choices have the greatest influence on this outcome?

1.3 Research Gap

Building on the problem statement above, four specific gaps can be identified in the existing literature.

First, **the lack of a controlled, side-by-side comparison**. Most published studies evaluate a single method, or at most two closely related variants of one method, rather than placing several structurally distinct families -- DWT thresholding, EMD-DWT, VMD-DWT and non-local means -- on an equal footing [2], [17], [18], [19].

Second, **limited attention to compound, simultaneous noise**. Clinical ECG signals are rarely contaminated by a single noise source, yet a large share of the denoising literature reports method performance against one noise type at a time, and the joint behaviour of methods under compound contamination remains less studied [3], [4].

Third, **incomplete reporting of morphology preservation**. A method can increase the signal-to-noise ratio dramatically while still distorting the QRS complex, the P wave or the T wave in clinically relevant ways; several studies report only aggregate error metrics, without a qualitative examination of these morphological landmarks [5], [18].

Fourth, **scarce evidence on the sensitivity of hybrid methods to wavelet configuration**. How the mother-wavelet family (e.g., Daubechies versus Symlet) and the decomposition level interact with the choice of decomposition front end (EMD versus VMD) has received relatively little systematic attention, although evidence from related studies suggests that this configuration matters [10], [11].

Addressing even a limited subset of these four gaps -- specifically the first, the third in quantitative form, and the fourth -- requires a dedicated comparative design, which is exactly what the methodology developed in Chapter 3 provides.

1.4 Necessity and Significance of the Research

The value of a controlled comparative evaluation is not limited to a single audience. For biomedical engineers and researchers, a common benchmark reduces the risk of adopting a denoising stage solely on the basis of a favourable but non-comparable result, and clarifies which of the design choices examined in this research -- decomposition front end, mother wavelet or decomposition level -- actually drives performance rather than merely correlating with it. For manufacturers of ECG recording equipment and portable cardiac monitors, an explicit picture of the trade-off between noise suppression and morphology preservation bears directly on firmware-level signal-conditioning decisions. For cardiologists, the ultimate beneficiaries of any improvement in signal quality are the diagnostic judgements that rely on an undistorted QRS complex and T wave and, indirectly, the patients whose arrhythmias or ischaemic changes are read from those same judgements.

From an academic perspective, this research also offers a reproducible protocol -- a fixed database subset, a fixed noise model and a fixed pair of evaluation metrics -- that subsequent comparative work in this field can adopt or extend, which in itself partly remedies the non-comparability problem described in the previous section.

1.5 Research Objectives

1.5.1 Main Objective

To provide a comparative evaluation of wavelet-based denoising methods for the electrocardiogram signal under identical experimental conditions.

1.5.2 Secondary Objectives

- 1- To compare a hybrid method that combines the wavelet transform with empirical mode decomposition (EMD-DWT) against a hybrid method that combines the wavelet transform with variational mode decomposition (VMD-DWT) in removing additive Gaussian noise from the ECG signal.
- 2- To investigate the effect of the mother-wavelet type (Daubechies-4 versus Symlet-4) and the decomposition level (levels 1 to 3) on the denoising quality of these hybrid methods.
- 3- To compare the best hybrid wavelet method with the non-local means method and to provide practical guidance for selecting an appropriate method.

1.6 Research Questions

Main question: Which wavelet-based denoising method performs best in removing noise from the electrocardiogram signal, and which factors influence the quality of this performance?

Secondary questions:

- 1- Does the hybrid wavelet transform with variational mode decomposition perform better in Gaussian denoising than the hybrid wavelet transform with empirical mode decomposition?
- 2- What effect do the mother-wavelet type and the decomposition level have on denoising quality?
- 3- Can wavelet-based methods match the performance of the non-local means method?

1.7 Research Hypotheses

Main hypothesis: Hybrid methods based on the wavelet transform combined with variational mode decomposition perform better in denoising the electrocardiogram signal than hybrid methods based on the wavelet transform combined with empirical mode decomposition, and an appropriate choice of mother wavelet and decomposition level improves denoising quality.

Secondary hypotheses:

- 1- [H1:] VMD-DWT provides a higher signal-to-noise ratio and a lower mean squared error than EMD-DWT in removing Gaussian noise from the ECG signal.
- 2- [H2:] The mother-wavelet type and the decomposition level have a significant effect on denoising quality, such that the Symlet-4 wavelet at higher decomposition levels performs better than other configurations.
- 3- [H3:] VMD-based hybrid wavelet methods provide performance comparable to the non-local means method and, under some noise conditions, even perform better.

1.8 Scope and Limitations of the Research

This research has been deliberately bounded in several respects, and these limitations should be regarded as intentional delimitations rather than accidental shortcomings. The test signals are drawn exclusively from the MIT-BIH Arrhythmia Database [20], [21]. The noise model is restricted to additive Gaussian noise at two variance levels (10 and 20); although

real, recorded artifacts such as power-line interference, baseline wander and muscle noise are clinically important in the wider literature [3], [4], their investigation is left to future work. The comparison itself is likewise limited to four denoising strategies -- DWT thresholding, EMD-DWT, VMD-DWT and non-local means -- using two mother wavelets (Daubechies-4 and Symlet-4) and three decomposition levels (1 to 3). Evaluation is quantitative, using the signal-to-noise ratio, the mean squared error and the preservation of QRS-complex amplitude. The entire implementation and analysis were carried out in MATLAB.

1.9 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the theoretical foundations of wavelet-based ECG denoising and places this research in the context of the empirical literature on DWT, EMD-DWT, VMD-DWT and the non-local means method. Chapter 3 describes in full detail the research design, the database, the noise model, the four denoising methods compared, the experimental variables and the evaluation metrics. Chapter 4 presents the implementation and the resulting quantitative and qualitative results. Chapter 5 summarizes these findings as final conclusions and sets out practical recommendations and directions for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter situates the present research within the body of work on wavelet-based denoising of the electrocardiogram signal. Section 2.2 begins with the theoretical foundations on which the subsequent empirical work rests: the structure of the ECG signal and the noise sources that contaminate it, the wavelet transform and wavelet thresholding, the two signal-adaptive decomposition paradigms examined in this thesis -- empirical mode decomposition and variational mode decomposition -- and the non-local means filter, which serves in this research as the non-wavelet reference method. Section 2.3 then reviews the related empirical literature, organized by methodological family rather than by chronological order of publication, because the aim of this review is to reveal how these families relate to one another, not to narrate the history of the field. Section 2.4 condenses the reviewed studies into a single comparative table, Section 2.5 critically synthesizes this material against the research gaps raised in Chapter 1, and Section 2.6 concludes the chapter.

2.2 Theoretical Background

2.2.1 The Electrocardiogram Signal and Its Noise Sources

A single cardiac cycle recorded at the body surface produces a characteristic sequence of deflections, conventionally labelled P, Q, R, S and T, corresponding respectively to atrial depolarization, the onset of ventricular depolarization, the peak of ventricular depolarization, its terminal phase, and ventricular repolarization [1], [2]. Diagnostic reasoning about arrhythmia, conduction disturbance and ischaemia rests almost entirely on the timing, amplitude and shape of these deflections, and for this reason the effect of a denoising method on signal morphology cannot be treated as secondary to its effect on aggregate error [2].

The contaminating sources fall into a small number of well-known categories. Power-line interference appears as a narrowband component at the mains frequency and its harmonics; baseline wander is a slow drift below 0.5 Hz caused by respiration, changes in skin-electrode impedance and patient movement; electromyographic noise from nearby skeletal muscles is broadband and can overlap substantially with the QRS spectrum; and motion artifacts,

caused by electrode displacement, are transient and can locally mimic a P wave or a small QRS complex, which makes them particularly harmful to automated diagnosis [3], [4]. Because these sources differ in bandwidth and statistical character, no single fixed filter response can remove all of them without degrading part of the signal itself [5], which is the structural reason why wavelet- and decomposition-based methods have replaced simple filtering in this field.

From a signal-processing perspective, it is useful to separate the effect of a noise source from its cause. Power-line interference and baseline wander are, in principle, the easiest to characterize, because their frequency content is narrow and largely predictable; nevertheless, a notch filter aggressive enough to remove 50 Hz interference can distort the T wave of a patient with a low heart rate whose energy extends into the same band [1]. Electromyographic and motion noise are far more difficult, because their spectral content overlaps directly with the QRS complex rather than lying in a separate band, so any attenuation sufficient to suppress them necessarily also attenuates part of the signal of interest [4]. Additive white Gaussian noise -- the idealized noise model used for the empirical work of Chapter 3 -- is analytically convenient precisely because it is spectrally flat and statistically fully specified, and for this reason it remains the standard first test case even in studies whose ultimate target is a more clinically realistic noise mixture [3], [17].

2.2.2 Wavelet Transform Theory

The discrete wavelet transform (DWT) decomposes a signal into a set of coefficients associated with scaled and translated versions of a single prototype function, the mother wavelet, and its associated scaling function [6]. Unlike the short-time Fourier transform, which fixes the time-frequency resolution trade-off through a constant window length, the resolution of the wavelet transform varies with scale: coarse scales, associated with low frequencies, are well resolved in frequency but coarsely in time, whereas fine scales, associated with high frequencies, are well resolved in time but coarsely in frequency [6]. This variable resolution is a good match for the ECG, whose diagnostically important content -- the short, high-frequency QRS complex and the slower, lower-frequency P and T waves -- is spread over different regions of the time-frequency plane within a single beat [5].

Using Mallat's fast pyramid algorithm, the DWT recursively splits the signal into approximation and detail coefficients through a pair of quadrature mirror filters, halving the sampling rate at each decomposition level [6]. The choice of mother wavelet -- in this thesis,

Daubechies-4 or Symlet-4 -- determines the shape of the analysing function and hence how efficiently a given morphological feature is represented by a small number of large coefficients, while the decomposition level determines how many times this splitting is repeated and hence which frequency bands are examined explicitly. Daubechies and Symlet wavelets are both compactly supported and orthogonal, which keeps the DWT computationally efficient and guarantees exact reconstruction in the absence of thresholding, but the two differ in a property that matters for ECG work: the Daubechies family is noticeably asymmetric, whereas the Symlet family is a nearly symmetric modification of the same construction, designed to reduce the phase distortion caused by asymmetry [11]. Because the QRS complex is itself a strongly asymmetric transient, the interaction between the intrinsic symmetry of the wavelet and the asymmetry of the signal is not clear in advance, and this is the main reason why the choice between Daubechies-4 and Symlet-4 is treated in this thesis as an independent variable rather than a fixed assumption.

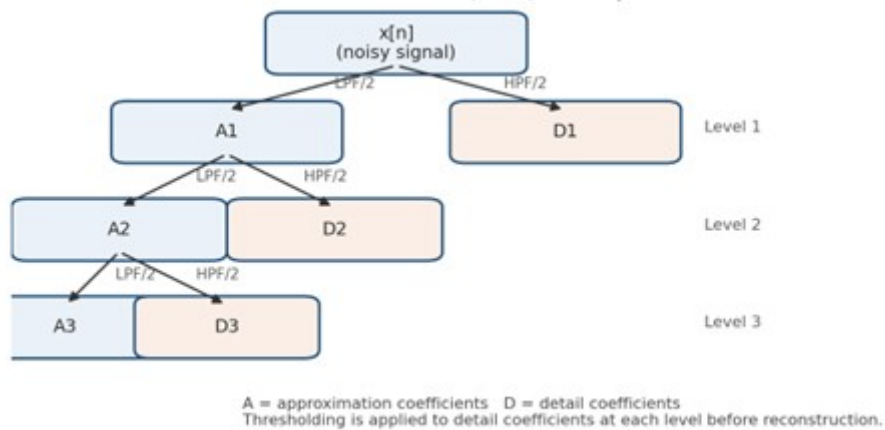


Figure 2.1: Three-level discrete wavelet transform decomposition tree (original illustration of the algorithm formulation in [6]).

2.2.3 Wavelet Thresholding

Wavelet thresholding denoises a signal by exploiting the fact that a well-chosen wavelet basis concentrates the energy of a clean signal in a small number of large coefficients, whereas additive noise spreads its energy almost uniformly across all coefficients [7], [8], [9]. After decomposition, coefficients below a threshold are regarded as noise-dominated and are set to zero (or shrunk towards zero), and the signal is then reconstructed from the remaining coefficients through the inverse transform [14]. The soft and hard thresholding rules of Donoho and Johnstone, together with the universal threshold derived from the noise standard deviation and the number of samples, remain the reference against which most

subsequent refinements are measured [7], [8]; Donoho and Johnstone's later work on adapting the threshold to unknown smoothness formally shows why a single, data-driven threshold can achieve performance close to that of the best possible threshold chosen with hindsight [9]. Soft thresholding shrinks the retained coefficients continuously towards zero and tends to produce a smoother reconstruction, whereas hard thresholding keeps coefficients above the threshold unchanged and can better preserve sharp features such as the QRS peak, at the cost of occasional discontinuities [14]. The practical quality of the reconstruction depends jointly on the thresholding rule, the threshold value, the mother wavelet and the decomposition level, which is exactly the set of design choices that this thesis holds under experimental control.

The threshold value is usually derived from an estimate of the noise standard deviation, most commonly computed from the median absolute value of the finest-level detail coefficients, combined with a rule such as Donoho and Johnstone's universal threshold or a level-dependent variant of it [7], [9]. A threshold set too low retains residual noise in the reconstruction; a threshold set too high removes low-amplitude but diagnostically relevant detail, such as the terminal portion of the T wave. This bias-variance trade-off also arises, in a different form, in the level-dependent thresholding schemes reviewed below, and it appears in exactly the same way whether thresholding is applied directly to the DWT coefficients of the raw signal or -- as happens in the two hybrid methods central to this thesis -- to the DWT coefficients computed separately on each intrinsic or variational mode.

2.2.4 Empirical Mode Decomposition

Empirical mode decomposition (EMD), introduced by Huang et al., decomposes a signal adaptively, without a predefined basis, into a finite set of intrinsic mode functions (IMFs) through an iterative "sifting" process that identifies local extrema, constructs upper and lower envelopes, and subtracts their local mean until the residue satisfies the formal conditions of an IMF [12]. Because the decomposition is derived from the signal itself rather than from a fixed set of basis functions, EMD adapts naturally to the non-stationary and non-linear nature of biological recordings, and each resulting IMF captures oscillations confined to a relatively narrow, local frequency band [12]. In ECG denoising, this property is exploited by treating the higher-order IMFs -- associated with fast, noise-dominated oscillations -- differently from the lower-order IMFs in which most of the signal energy resides, and by applying wavelet thresholding to some or all IMFs before recombining them

into the denoised signal, an approach examined systematically by Lahmiri [14] and by Kabir and Shahnaz [15]. The main known limitation of EMD, mode mixing -- in which a single IMF contains oscillations of disparate frequency, or a single frequency is split across adjacent IMFs -- is one of the main motivations for the variational alternative examined next [13].

The EMD sifting process also lacks a stopping rule derived from first principles; in practice, sifting stops once a standard-deviation-based criterion between successive iterations falls below a fixed tolerance, a pragmatic choice that leaves the number of IMFs and their exact shape somewhat sensitive to the tolerance parameter and to boundary effects at the start and end of the signal [12]. These well-documented sensitivities are part of the motivation for examining an ensemble refinement such as CEEMDAN in Section 2.3.3, as well as for including a fully non-recursive alternative, VMD, in the direct comparison of this thesis.

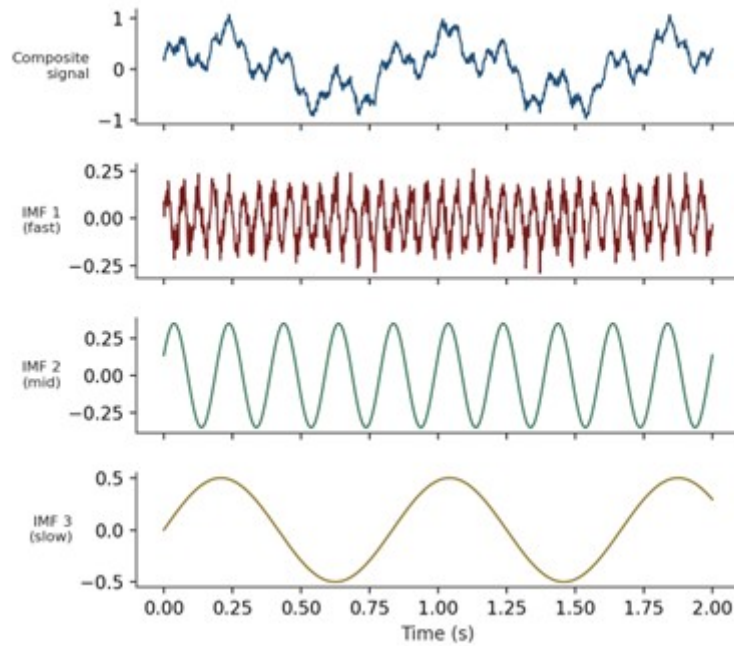


Figure 2.2: Schematic illustration of empirical mode decomposition applied to a synthetic composite signal, showing the progressive separation into intrinsic modes of decreasing frequency (original illustration of the algorithm described in [12]).

2.2.5 Variational Mode Decomposition

Variational mode decomposition (VMD), proposed by Dragomiretskiy and Zosso as a non-recursive alternative to EMD, formulates the decomposition as a constrained variational optimization problem: it seeks a fixed number of band-limited modes, each demodulated to

baseband through a Hilbert-transform-based analytic signal construction, such that their sum reconstructs the original signal while the bandwidth of each mode, penalized in the objective function, is minimized [13]. Because the number of modes is specified in advance and the decomposition is solved globally rather than through iterative sifting, VMD is noticeably less prone to the mode-mixing artifacts that affect EMD, although at the cost that the number of modes -- and, in some formulations, the bandwidth-constraint parameter -- must be set in advance [13]. In the hybrid VMD-DWT configuration examined in this thesis, wavelet thresholding is applied to each variational mode in exactly the same way as it is applied to each IMF in the EMD-DWT configuration, so any performance difference measured between the two hybrid methods can be attributed to the decomposition front end rather than to a difference in the thresholding stage itself [14].

The variational problem underlying VMD is solved iteratively using the alternating direction method of multipliers, which updates in turn the spectrum of each mode, its centre frequency and a Lagrange multiplier, in contrast to the local, sample-by-sample envelope construction on which EMD relies [13]. This global, frequency-domain formulation is the reason for VMD's improved robustness to noise and reduced mode mixing relative to EMD, but it means that the number of modes to be extracted must be specified before the decomposition begins, rather than emerging automatically from the signal as in EMD; in this thesis, following the underlying research proposal, this parameter is held fixed at five modes for both decompositions, so that the two front ends are compared under identical mode-count conditions.

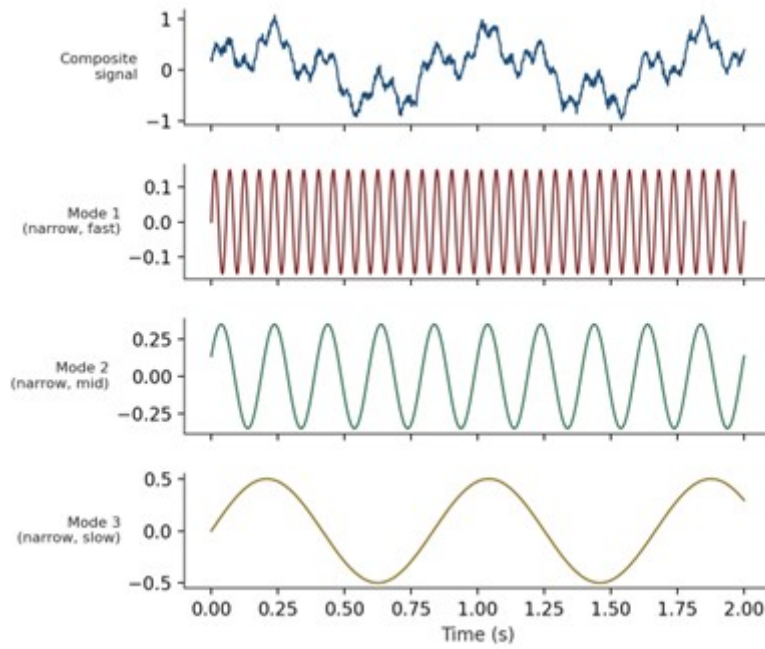


Figure 2.3: Schematic illustration of variational mode decomposition applied to the same synthetic composite signal as in Figure 2.2, showing the narrower-band separation of the modes (original illustration of the algorithm described in [13]).

2.2.6 Non-Local Means Filtering

The non-local means (NLM) filter, introduced by Buades, Coll and Morel for denoising natural images, estimates each sample of a signal as a weighted average of other samples, where the weight assigned to a candidate sample reflects the similarity between the local neighbourhood (patch) around the sample being estimated and the local neighbourhood around each candidate sample, rather than mere spatial or temporal proximity [16]. Applying this idea to a one-dimensional quasi-periodic signal such as the ECG is a natural fit: successive cardiac cycles repeat with similar but not identical morphology, so a patch from one heartbeat can meaningfully inform the denoised estimate of the corresponding patch in a neighbouring beat, even if the two patches are separated in time by a full cycle [16]. Unlike EMD-DWT and VMD-DWT, NLM performs no explicit frequency-domain decomposition and requires no choice of mother wavelet or decomposition level, which is precisely why it is used in this thesis as an independent conceptual reference against the hybrid wavelet methods (third research question, Section 1.6).

The estimation power of NLM for any given sample depends mainly on two parameters: the size of the local patch used to judge similarity, and a filtering parameter that determines how strongly the weighting is pulled towards highly similar patches (as opposed to moderately similar ones). A patch that is too short may spuriously match unrelated parts of

the signal, whereas a patch that is too long may fail to find sufficiently similar patches in a recording of finite length; the filtering parameter plays a role similar to that of the threshold in wavelet thresholding, balancing noise suppression against the risk of over-smoothing genuine beat-to-beat variability [16]. Because NLM makes no assumption about frequency content, it is in principle insensitive to the particular noise sources discussed in Section 2.2.1, which is part of the reason it is useful as a reference point rather than as a competitor confined to the same design space as the two hybrid wavelet methods.

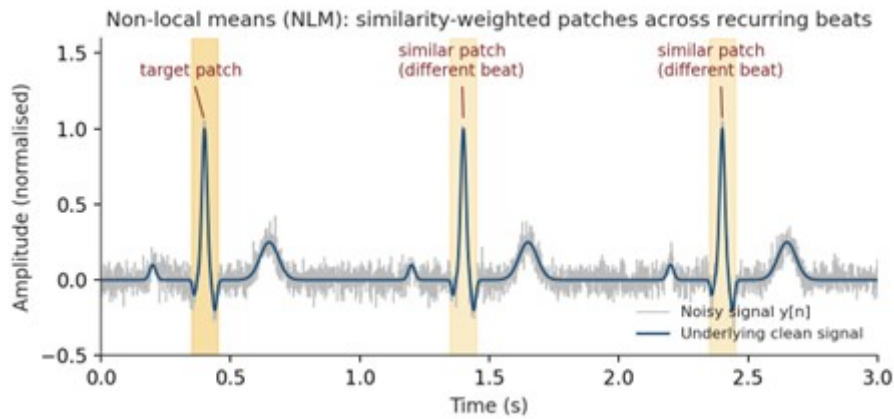


Figure 2.4: Conceptual illustration of non-local means patch similarity across quasi-periodic ECG beats (original illustration of the algorithm described in [16]).

2.3 Review of Previous Research on Wavelet-Based ECG Denoising

2.3.1 Classical DWT Thresholding Methods

Alfaouri and Daqrouq were among the first groups to apply wavelet-coefficient thresholding systematically to the ECG signal, reporting that properly tuned thresholds improve the signal-to-noise ratio without noticeable distortion of the QRS complex [22]. Poornachandra extended this line of research with a subband-dependent threshold that adapts to the noise level estimated at each decomposition level rather than applying a single universal threshold, an approach that generally outperforms a fixed universal threshold when the ECG noise floor varies across frequency bands [10]. Singh and Tiwari addressed a complementary and different question -- which mother wavelet is fundamentally best suited to ECG morphology -- and showed that the choice of wavelet basis, independently of the thresholding rule subsequently applied, noticeably affects reconstruction fidelity [11], a finding directly related to the second secondary objective of this thesis. More recently, Kamal et al. combined a discrete wavelet transform architecture with careful selection of the threshold and decomposition level, reporting a correlation coefficient above 0.95 and a

signal-to-noise ratio improvement over earlier baselines of roughly one to two decibels under power-line interference, baseline wander and Gaussian noise [17], results that show how much incremental improvement can still be extracted from the classical DWT thresholding family through careful tuning alone.

2.3.2 Adaptive and Stationary Wavelet Methods

A second group of studies departs from the plain, critically sampled DWT towards shift-invariant or total-variation-regularized variants. Madan et al. proposed a weighted stationary wavelet transform combined with total-variation minimization, arguing that the shift invariance of the stationary wavelet transform prevents the pseudo-Gibbs artifacts that a critically sampled DWT can produce near sharp transitions such as the QRS complex, and reported effective suppression of several clinically relevant noise types [5]. Rezgui, Nasraoui and Talbi pursued a similar idea through a convex fused-lasso formulation combined with non-convex regularization and total variation, reporting a favourable signal-to-noise ratio relative to simpler convex baselines [23]. Malik et al. combined iterative filtering with a lifting-scheme wavelet transform to remove power-line interference and baseline wander simultaneously, reporting that the lifting-based implementation offers a favourable trade-off between computational cost and denoising quality, suitable for embedded acquisition hardware [3]. Taken together, this group shows that a considerable part of the improvement attributed in the literature to "wavelet-based denoising" is in fact attributable to the particular transform variant and regularization scheme used rather than to wavelet thresholding per se -- a distinction that this thesis keeps explicit by holding the thresholding stage fixed across its hybrid methods.

2.3.3 Hybrid EMD/VMD-Wavelet Methods

The methodological family most directly related to this thesis combines a signal-adaptive decomposition front end with wavelet thresholding applied to the resulting modes. Lahmiri's comparative study is the closest methodological predecessor: by applying wavelet thresholding in both the EMD domain and the VMD domain under identical conditions, it showed that thresholding in the VMD domain competes with, and in some configurations is preferable to, thresholding in the EMD domain, although it cautioned that this result depends on the particular wavelet and thresholding rule used in the second stage [14]. Kabir and Shahnaz combined EMD with a noise-reduction algorithm operating jointly in the EMD and wavelet domains and reported improved performance over single-domain thresholding,

attributing the improvement to the complementary capabilities of the two decompositions -- the adaptability of EMD to non-stationarity and the ability of the wavelet transform to localize and shrink the residual noise energy within each mode [15]. Ge et al. combined an improved complete ensemble EMD with adaptive noise (CEEMDAN) with wavelet thresholding and singular spectrum analysis in a three-stage pipeline, reporting a 39 to 55 percent signal-to-noise ratio improvement over earlier CEEMDAN-wavelet baselines, which shows both the promise and the growing structural complexity of this hybrid family [4]. Nassiri, Nahraoui and Elouaham applied a related wavelet-based hybrid method to the joint problem of ECG compression and denoising, showing that a favourable compression ratio and preservation of signal quality are not necessarily conflicting goals when the decomposition stage is chosen carefully [24]. None of these hybrid studies benchmarks its EMD or VMD pipeline against a non-decomposition reference method, which is precisely one of the specific comparisons that this thesis undertakes (third secondary objective, Section 1.5.2).

2.3.4 Deep-Learning and Generative Methods: A Point of Comparison

A separate and increasingly active branch of the literature replaces or augments the wavelet-thresholding stage with a learned model. Chatterjee et al. combined a sparsity-based, Huber-penalized optimization stage for baseline-wander removal with wavelet-domain thresholding and an extreme-learning-machine stage for residual noise, reporting a signal-to-noise ratio improvement of up to 27.87 dB on record 119 of the MIT-BIH database against five noise types, including compound noise [19] -- the largest single improvement reported among the studies reviewed here, although with a far more complex, multi-stage pipeline than the DWT, EMD-DWT or VMD-DWT methods examined in this thesis. Maulida et al. combined wavelet-domain features with an improved filter bank for arrhythmia classification rather than for denoising alone, showing that wavelet decomposition continues to serve as a feature-extraction front end even when the downstream task is classification rather than signal reconstruction [25]. Esmaili Alidash and Danandeh Hesar applied a convolutional neural network directly to abdominal ECG recordings for fetal QRS-complex detection, without an explicit denoising stage, illustrating the alternative strategy of learning robustness to noise instead of removing it beforehand [26]. Maghsoudi Gombavani, Fadaei Eslam and Yaghmaei applied a wavelet-transform-based generative adversarial network to denoising in the image domain, a result outside the ECG field but methodologically related,

since it shows that the wavelet transform still acts as a structural prior even within deep generative architectures [27]. These studies are reviewed here as a point of comparison rather than as directly comparable work, because this thesis is concerned specifically with classical, non-learned hybrid wavelet methods and their standing relative to non-local means.

2.3.5 Related Applications in Neighbouring Fields

Evidence for the general applicability of wavelet- and similarity-based decomposition also comes from outside the ECG literature. Vaghefi et al. applied the wavelet packet transform to acoustic signals for the non-contact fault diagnosis of polymer gears, extracting statistical features from the fourth-level coefficients, and showed that several standard classifiers can reliably distinguish healthy from faulty gears [28], a result that supports the more general claim that wavelet decomposition concentrates diagnostically relevant information in a manageable number of coefficients, independently of the particular signal domain. Rahemi, Sharii and Mousavi Mirklaei combined the wavelet transform with a type-2 fuzzy system to improve weak-signal acquisition in GPS receivers, again exploiting the time-frequency localization of the transform to recover structure under low signal-to-noise-ratio conditions [29]. Yousefvand, Sepahvand and Bibak, working directly on ECG denoising but from an adaptive-filtering rather than a wavelet perspective, proposed an improved affine projection algorithm that reduced the computational load by more than 40 percent relative to common adaptive filters while maintaining denoising performance, an emphasis that computational efficiency -- one of the research gaps identified in Section 1.3 -- remains a live concern even for methods outside the wavelet family [30]. The non-linear Bayesian filtering framework of Sameni et al., based on an explicit dynamical model of the ECG rather than on wavelet decomposition, provides another reference point for how much can be achieved with model-based rather than transform-based denoising [31].

2.4 Summary Table of the Reviewed Studies

Table 2.1 condenses the ten studies most directly related to the present comparison and lists, for each, the method family, the database used, the noise type(s) examined and the main quantitative result reported by the original authors. The table is included to make explicit -- at a glance -- the heterogeneity described qualitatively above: no two rows share the same noise model and evaluation protocol, which is the empirical basis of the research gap raised in Section 1.3.

Table 2.1: Summary of representative wavelet-based ECG denoising studies reviewed in this chapter.

Study	Method	Database	Noise type(s)	Key reported result
Kamal et al. [17]	DWT thresholding architecture	MIT-BIH + synthetic	Power-line interference, baseline wander, WGN	SNR improvement up to 17.97 dB; correlation up to 0.9934
Yang and Wei [18]	Morphologically stable wavelet thresholding	MIT-BIH	Compound	MSE 0.0146; SNR 68.69 dB; QRS amplitude preserved
Chatterjee et al. [19]	Sparsity-based wavelet denoising autoencoder	MIT-BIH	BW, WGN, EMG, PLI, compound	SNR improvement up to 27.87 dB (record 119)
Madan et al. [5]	Weighted stationary-wavelet total variation	MIT-BIH	Compound clinical noise	Effective removal of several noise types
Malik et al. [3]	Iterative filtering + lifting wavelet	MIT-BIH	Power-line interference, baseline wander	Improved coefficient-thresholding performance
Ge et al. [4]	Improved CEEMDAN + wavelet thresholding + SSA	MIT-BIH	Compound noise	39-55% SNR improvement over earlier baselines
Rezgui et al. [23]	Fused-lasso wavelet/total-variation denoising	MIT-BIH	Compound	Favourable SNR relative to convex baselines
Lahmiri [14]	Wavelet thresholding in the EMD vs. VMD domain	MIT-BIH	Synthetic	VMD-domain thresholding competitive with EMD domain
Nassiri et al. [24]	Compression + wavelet-based denoising	MIT-BIH	Compound	Balanced compression ratio and quality preservation
Sameni et al. [31]	Non-linear Bayesian filtering framework	MIT-BIH, synthetic	WGN, ectopic beats	Robust tracking of the ECG dynamical model

2.5 Critical Synthesis and Identified Research Gaps

Reading across the three method families reviewed in Section 2.3, a consistent pattern emerges. Classical DWT thresholding (Section 2.3.1) remains a strong and computationally inexpensive baseline whose performance is highly sensitive to the thresholding rule, the mother wavelet and the decomposition level -- exactly the parameters that this thesis holds under experimental control. Adaptive and stationary wavelet variants (Section 2.3.2) generally improve on the plain DWT by trading increased computational cost for shift invariance or added regularization, but at the cost of making it harder to disentangle whether an improvement stems from the transform variant or from the thresholding rule. Hybrid EMD/VMD-wavelet methods (Section 2.3.3) promise further improvement by adapting the decomposition itself to the signal, and Lahmiri's direct comparison of EMD versus VMD is the strongest available evidence that the choice between the two matters [14]; however, that comparison, like almost every other study reviewed here, does not extend to a non-wavelet reference method, which leaves open a question that this thesis addresses directly through the inclusion of non-local means (Section 2.2.6).

Three of the four research gaps raised in Section 1.3 recur concretely throughout this review. The lack of a controlled, side-by-side comparison (first gap) is evident in the fact that no two studies in Table 2.1 share the same noise model. Incomplete reporting of morphology

preservation (third gap) is evident in the fact that only Yang and Wei [18] and Chatterjee et al. [19] report explicit evidence of QRS preservation alongside their aggregate error metrics, while the remaining studies report only SNR or MSE. Scarce evidence on the sensitivity of the wavelet configuration within hybrid methods (fourth gap) is evident in the fact that the wavelet-selection findings of Singh and Tiwari [11] and the subband-threshold findings of Poornachandra [10] were established for single-stage DWT thresholding and have not been revisited within an EMD or VMD hybrid method. Together, these three gaps define the methodological requirements that the experimental design of Chapter 3 must satisfy: a single database subset, a single noise model applied identically to all four methods, explicit variation of the mother wavelet and decomposition level within the hybrid methods, and a morphology-focused evaluation alongside the quantitative SNR and MSE metrics.

It should be noted that this synthesis does not claim that any individual study reviewed above is methodologically flawed; each is internally consistent and reports results that are reliable under its own conditions. The claim is narrower but, for the purposes of this thesis, more important: that the collection of these studies, as a body of evidence, does not by itself answer the question a practitioner actually needs answered -- which method to choose, under which conditions, for a given application -- because the studies do not share a common protocol against which their answers can be directly measured. A single controlled comparative study, even of limited scope such as this thesis, therefore complements this literature rather than repeating it, and fills a specific evidential gap that no number of additional single-method studies can close on their own.

2.6 Chapter Summary

This chapter reviewed the theoretical foundations of wavelet-based ECG denoising, from the wavelet transform and wavelet thresholding to the two signal-adaptive decompositions -- empirical mode decomposition and variational mode decomposition -- that underlie the two hybrid methods compared in this thesis, and the non-local means filter used here as an independent reference. The subsequent empirical review revealed a field rich in individually credible results but weak in mutual comparability, a pattern documented concretely in Table 2.1 and traced to the specific research gaps identified in Chapter 1. Chapter 3 now turns these gaps into a concrete experimental design.

Chapter 3

Research Methodology

3.1 Research Design

This research is a quantitative, simulation-based, comparative-experimental design implemented entirely in MATLAB. Rather than proposing a new denoising algorithm, the design holds a set of experimental factors fixed across four existing denoising strategies, so that observed differences in outcome can be attributed to the strategies themselves and their internal design parameters rather than to differences in data, noise or evaluation protocol. This choice follows directly from the research gap raised in Chapter 1: because the wider literature rarely holds these factors fixed across the methods it compares (Section 2.5), the contribution of this thesis lies largely in the experimental control described in this chapter, and not merely in the methods themselves.

The independent variables are the denoising method (DWT, EMD-DWT, VMD-DWT or NLM), the mother wavelet used in the wavelet-thresholding stage (Daubechies-4 or Symlet-4), and the DWT decomposition level (1, 2 or 3). The dependent variables are the signal-to-noise ratio, the mean squared error and the preservation of QRS-complex amplitude, all computed between each denoised signal and its known, clean original. A specific set of control variables -- described in Section 3.6 and summarized in Table 3.1 -- is held fixed throughout the experiment, so that no observed difference between methods can be attributed to an uncontrolled change in the data or in the noise itself.

3.2 Database Description

The test signals are drawn from the MIT-BIH Arrhythmia Database, a collection of 48 half-hour, two-channel ambulatory ECG recordings sampled at 360 Hz, annotated by independent cardiologists and maintained on PhysioNet [20], [21]. The extensive and long-standing use of this database as a de facto standard in ECG signal-processing research is precisely what makes it suitable for this study: because a large body of previous work, including several studies reviewed in Chapter 2 [3], [4], [17], [18], [19], [14], reports its results on this same corpus, reusing it keeps the present comparison interpretable against, and in principle extensible to, that wider literature.

A total of twenty records were used (records 100 to 109, 111 to 119, and 121). Of these, one record (record 100) was set aside exclusively for calibrating the VMD bandwidth-penalty parameter (Section 3.4.4) and was excluded from all reported evaluation results, so that this parameter is never tuned on the same data used to score the methods. The remaining nineteen records form the evaluation set. This sample size was deliberately extended beyond the five records specified in the original research proposal, an extension adopted to provide a stronger and more reliable statistical evaluation of the methods; Chapters 3 and 4 therefore both describe this extended twenty-record set consistently. Record selection followed a principled, reproducible procedure rather than a purely random choice: a list of twenty standard MIT-BIH record identifiers was compiled, and each candidate was verified for retrievability from PhysioNet before inclusion. Because the same fixed set of clean signals, and the same noisy signal generated from each, is used identically across all four denoising methods (Section 3.5), neither this extension of the sample size nor this selection procedure places any method at an advantage or disadvantage relative to the others.

For each record, 1000 consecutive samples (after skipping an initial transient) were extracted in raw digitized units (ADC), and the sample mean of each record was subtracted from it to remove the DC offset, since a constant offset is not the target of any of the four denoising methods and would otherwise dominate the signal-power term in the SNR formula without reflecting actual denoising performance.

3.3 Noise Model

Each clean reference signal is contaminated with additive Gaussian noise, where s is the clean signal, n is a zero-mean Gaussian random process, and y is the resulting noisy signal presented to each denoising method:

$$(3.1)$$

Two noise variances were used, 10 and 20, so that each method is evaluated under both a moderate and a more severe contamination level. For each (record, variance) combination, three independent noise realizations were generated and evaluated separately, so that the reported results reflect performance over several random draws of noise rather than a single, potentially favourable or unfavourable instance. As stated in the limitations of Section 1.8, this thesis restricts its noise model to additive Gaussian noise; real, recorded artifacts such as

the power-line, motion and muscle interference listed in Section 2.2.1 are recognized as clinically important but are deliberately left outside the present scope.

3.4 Denoising Methods Compared

3.4.1 Method 1: Discrete Wavelet Transform (DWT), the Baseline

The discrete wavelet transform thresholding procedure described in Section 2.2.3 is applied directly to the noisy signal, without any signal-adaptive decomposition front end: the signal is decomposed using the Daubechies-4 or Symlet-4 mother wavelet at decomposition level 1, 2 or 3, the detail coefficients at each level are soft-thresholded using the Donoho-Johnstone universal threshold [7], [8], [9], and the signal is reconstructed through the inverse transform. This method serves as the baseline against which the added value of a signal-adaptive decomposition front end (Methods 2 and 3) is measured, and it directly addresses the fairness requirement of isolating the decomposition stage as the only varying factor among DWT, EMD-DWT and VMD-DWT.

3.4.2 Method 2: Empirical Mode Decomposition with DWT (EMD-DWT)

The noisy signal is first decomposed by empirical mode decomposition into at most five intrinsic mode functions plus a residue, following the sifting procedure of Huang et al. [12]. The same wavelet-thresholding procedure used in Method 1 is then applied independently to each intrinsic mode and to the residue, and the thresholded components are summed to reconstruct the final denoised signal, an approach consistent with the EMD-domain thresholding examined by Lahmiri [14] and by Kabir and Shahnaz [15].

3.4.3 Method 3: Variational Mode Decomposition with DWT (VMD-DWT)

The noisy signal is instead decomposed by variational mode decomposition into five band-limited modes, obtained by solving the constrained variational optimization problem of Dragomiretskiy and Zosso [13]. The same wavelet-thresholding procedure is applied independently to each mode, and the thresholded modes are summed to reconstruct the final signal. Because Methods 2 and 3 share exactly the same wavelet-thresholding stage and differ only in the decomposition front end, any measured difference in SNR or MSE between them is attributable specifically to the choice of EMD versus VMD, which is the comparison called for by the first secondary objective (Section 1.5.2) and the first research question (Section 1.6).

3.4.4 Calibration of the VMD Bandwidth-Penalty Parameter

VMD requires a bandwidth-penalty parameter, α , which is not specified in the approved objectives and must therefore be chosen empirically. To avoid tuning this parameter on the same data used to evaluate the methods, record 100 was set aside exclusively for calibration and excluded from the evaluation set (Section 3.2). Eight candidate values (50, 100, 150, 200, 300, 500, 1000 and 2000) were each scored by the mean post-thresholding SNR they produce on the calibration record, using the Symlet-4 wavelet at decomposition level 2 as a representative configuration and averaging over both noise variances; the candidate with the highest calibration score was then held fixed for all subsequent VMD-DWT computations on the nineteen evaluation records.

3.4.5 Method 4: Non-Local Means (NLM), the Reference Method

The non-local means filter is applied directly to the noisy signal, without any decomposition or wavelet-thresholding stage, following the patch-similarity weighting scheme of Buades, Coll and Morel [16]. Each denoised sample is computed as a weighted average of samples drawn from across the signal, with weights determined by the similarity between the local patch around the sample being estimated and the local patch around each candidate sample. NLM is designated as the reference method in this design precisely because it shares no structural component with Methods 1 to 3, which makes it a meaningful external benchmark for the third research question (Section 1.6) and the third secondary objective (Section 1.5.2).

3.5 Ensuring a Fair, Side-by-Side Comparison

Five design choices keep the comparison among all four methods scientifically fair. First, all four methods are evaluated on the same set of nineteen evaluation records, using identical sample windows. Second, within a given trial, all four methods are evaluated against the same realization of additive Gaussian noise -- that is, the same noisy signal is presented to DWT, EMD-DWT, VMD-DWT and NLM -- so no method receives an easier or harder noise instance than another. Third, DWT, EMD-DWT and VMD-DWT all use an identical wavelet-thresholding procedure (the same threshold rule and the same soft-thresholding formula), applied to the raw signal, to the intrinsic modes or to the variational modes, thereby isolating the decomposition front end as the only varying factor among these three methods. Fourth, all four methods are scored with the same metrics (SNR, MSE and QRS amplitude

preservation), computed against the same known clean reference signal. Fifth, all four methods are implemented and run within the same MATLAB software environment, which removes platform/implementation differences as a possible confounding factor.

3.6 Experimental Variables

Table 3.1 summarizes the independent, dependent and control variables of the experimental design described above.

Table 3.1: Independent, dependent and control variables of the experimental design.

Variable type	Variable	Levels / values
Independent	Denoising method	DWT, EMD-DWT, VMD-DWT, non-local means (NLM)
Independent	Mother wavelet (in the DWT stage)	Daubechies-4 (db4), Symlet-4 (sym4)
Independent	DWT decomposition level	Level 1, level 2, level 3
Dependent	Signal-to-noise ratio (SNR)	Measured in decibels
Dependent	Mean squared error (MSE)	Measured on normalized amplitude
Dependent	QRS-complex amplitude preservation	Percentage ratio and absolute error relative to the clean signal
Control	Database	MIT-BIH Arrhythmia Database [20], [21]
Control	Signal length	1000 samples per record
Control	Noise type and level	Additive Gaussian noise, variance 10 and 20, three independent realizations each
Control	EMD/VMD number of modes	Fixed at 5 modes for both decompositions
Control	VMD bandwidth-penalty parameter	Alpha, selected by calibration on a held-out record (Section 3.4.4)

3.7 Evaluation Metrics

Three metrics are used to score each method/parameter configuration.

Signal-to-noise ratio (SNR). The SNR expresses the power of the useful signal relative to the residual error after denoising, in decibels, and is computed as follows:

$$(3.2)$$

where s is the original clean signal, $\hat{s}$ is the denoised signal produced by a given method, and N is the number of samples (in this research, $N = 1000$). Higher SNR values indicate better denoising performance.

Mean squared error (MSE). The MSE expresses the magnitude of the difference between the clean signal and the denoised signal, and is computed as follows:

$$(3.3)$$

Lower MSE values indicate better denoising performance.

QRS-complex amplitude preservation. R-peak locations are identified automatically on the clean reference signal using a standard peak-detection procedure (minimum height of 50 percent of the record's maximum amplitude, minimum inter-peak distance corresponding to the highest plausible physiological heart rate). For each denoised signal, the amplitude at each of these sample locations is compared with its counterpart in the clean signal, yielding a mean amplitude-preservation ratio and a mean absolute percentage error for each method. This metric provides a quantitative measure of the morphology-preservation concern raised in the third gap (Section 1.3).

3.8 Experimental Procedure

For each of the nineteen evaluation records and each of the two noise variances, three independent noise realizations are generated. For each realization, the noisy signal is decomposed once by EMD and once by VMD; DWT thresholding is then applied, with both mother wavelets and all three decomposition levels, to the raw noisy signal (Method 1), to each EMD intrinsic mode and the residue (Method 2), and to each VMD mode (Method 3); NLM is applied once (Method 4, with no wavelet/level parameters). This procedure yields a total of scored configurations (twenty configurations per trial: one no-denoising baseline, six DWT, six EMD-DWT, six VMD-DWT and one NLM). SNR and MSE are computed for every configuration, and QRS amplitude preservation is computed wherever R peaks are detected in the corresponding record.

3.9 Implementation Tools

All stages of signal acquisition, noise contamination, algorithm implementation, numerical computation and statistical analysis were carried out in MATLAB, using the `emd` and `vmd` functions of the Signal Processing Toolbox for the two decomposition front ends and its `findpeaks` function for R-peak detection, and the `pdist2` function of the Statistics and Machine Learning Toolbox for computing patch similarity in non-local means. The discrete wavelet transform thresholding stage was implemented directly from the published Daubechies-4 and Symlet-4 filter coefficients [6], and was validated against a self-contained perfect-reconstruction test before use. The records were retrieved directly from PhysioNet [20], [21]. Using a single software environment for every method and configuration removes

platform/implementation differences as a possible confounding factor, consistent with the emphasis on experimental control raised in Section 3.1.

3.10 Chapter Summary

This chapter turned the research gaps identified in Chapters 1 and 2 into a concrete, controlled experimental design: a fixed dataset of nineteen evaluation records (plus a twentieth record held out for parameter calibration) from the MIT-BIH Arrhythmia Database; a fixed additive Gaussian noise model with two variances repeated over three independent realizations; four denoising methods -- a plain DWT baseline, EMD-DWT, VMD-DWT and non-local means -- evaluated identically; explicit variation of the mother wavelet and decomposition level within the three wavelet-thresholding methods; and a common set of three quantitative metrics. This design directly operationalizes the three research questions and three hypotheses stated in Sections 1.6 and 1.7, and the fairness safeguards set out in Section 3.5 respond directly to the stated requirement that the comparison be scientifically fair. The implementation and resulting findings are presented in Chapter 4.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter reports the findings of the comparative experiment designed in Chapter 3 and interprets them against the research questions and hypotheses stated in Section 1.7. Four methods are compared: a discrete wavelet transform (DWT) baseline, EMD-DWT, VMD-DWT and non-local means (NLM). The experiment covers nineteen records from the MIT-BIH Arrhythmia Database, both noise variances, and three independent Gaussian realizations per (record, variance) combination, yielding 114 independent trials and 684 matched configurations for each of the three wavelet-thresholding methods (). Section 4.2 presents the quantitative SNR/MSE results together with paired statistical testing; Section 4.3 provides a quantitative morphological evaluation based on real QRS peaks; Section 4.4 discusses each hypothesis in turn; Section 4.5 summarizes the contributions of this chapter; Section 4.6 relates the findings to the literature of Chapter 2; Section 4.7 states the remaining limitations; and Section 4.9 documents the implementation sufficiently for it to be independently reproducible.

The bandwidth-penalty parameter of variational mode decomposition (α) was calibrated systematically as described in Section 3.4.4: record 100 was treated as a held-out calibration record and excluded from evaluation, and the value $\alpha = 200$ achieved the highest calibration score (22.92 dB, versus 21.77 dB for $\alpha = 2000$, the value more common in the literature for other signal types) and was held fixed for all subsequent computations on the remaining nineteen records.

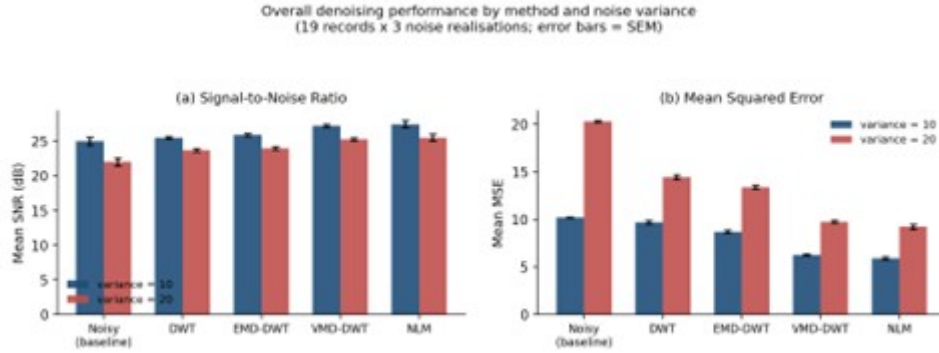
4.2 Quantitative Results

4.2.1 Overall Comparison of the Four Methods

Table 4.1 reports the overall mean SNR and MSE for each of the five conditions (no denoising, plus the four methods), averaged across all 19 records, both noise variances and all three noise realizations; for the three wavelet-thresholding methods, the average is additionally taken across both mother wavelets and all three decomposition levels.

Table 4.1: Overall mean SNR and MSE for the no-denoising baseline and each of the four methods.

Condition	Mean SNR (dB)	Mean MSE
No denoising (baseline)	23.51 4.57	15.25 5.10
DWT (baseline)	24.61 4.16	12.08 4.94
EMD-DWT	24.96 4.09	11.05 4.23
VMD-DWT	26.28 4.16	8.02 2.64
NLM (reference)	26.50 4.04	7.55 2.32

**Figure 4.1:** Overall comparison of SNR (a) and MSE (b) by method, broken down by noise variance.

Averaged across all configurations, NLM (26.50 dB) and VMD-DWT (26.28 dB) show the strongest performance and are close to each other, both clearly ahead of EMD-DWT (24.96 dB) and the DWT baseline (24.61 dB), which in turn improve on the no-denoising baseline (23.51 dB). The best single configuration observed in this entire study is VMD-DWT with the Symlet-4 wavelet at decomposition level 3 (26.55 dB, MSE = 7.56).

4.2.2 Effect of Noise Variance

Every method loses between 1.9 and 3.0 dB of SNR when the noise variance doubles from 10 to 20, which is the expected direction and a reasonable sanity check on the computations. The relative ordering of all five conditions remains unchanged at both noise levels.

4.2.3 Does Signal-Adaptive Decomposition Add Value over Plain Wavelet Thresholding?

Because the DWT baseline (Method 1) shares the same thresholding stage as EMD-DWT and VMD-DWT and differs from them only in the absence of a decomposition front end, the comparison between DWT and the two hybrid methods isolates the specific contribution of decomposing the signal before thresholding. Both hybrid methods significantly outperform the DWT baseline: EMD-DWT improves on DWT by a mean of 0.35 dB (, far below 0.001), and VMD-DWT improves on DWT by a mean of 1.68 dB (, far below 0.001). This result directly answers the concern of whether decomposition-based hybrid methods justify their

added implementation complexity: both add value relative to the simpler baseline, and the advantage of VMD over plain DWT is approximately 4.8 times larger than that of EMD.

4.2.4 Effect of the Mother-Wavelet Type

A paired comparison detects a statistically significant difference between Daubechies-4 and Symlet-4 for VMD-DWT (mean difference 0.05 dB, below 0.0001), in the direction hypothesized in Section 1.7 (Symlet-4 superior to Daubechies-4), but the corresponding values for DWT and EMD-DWT are not significant (and , respectively). In any case, the size of the wavelet effect -- at most one twentieth of a decibel -- is negligible compared with the 0.35-1.68 dB contributed by decomposition (Section 4.2.3) or the 0.66-2.36 dB contributed by the decomposition level (Section 4.2.5). This illustrates a more general statistical point: with a sufficiently large sample, even a practically unimportant difference can become detectable, and a statistical-significance result should be read alongside its effect size, not instead of it.

4.2.5 Effect of the DWT Decomposition Level

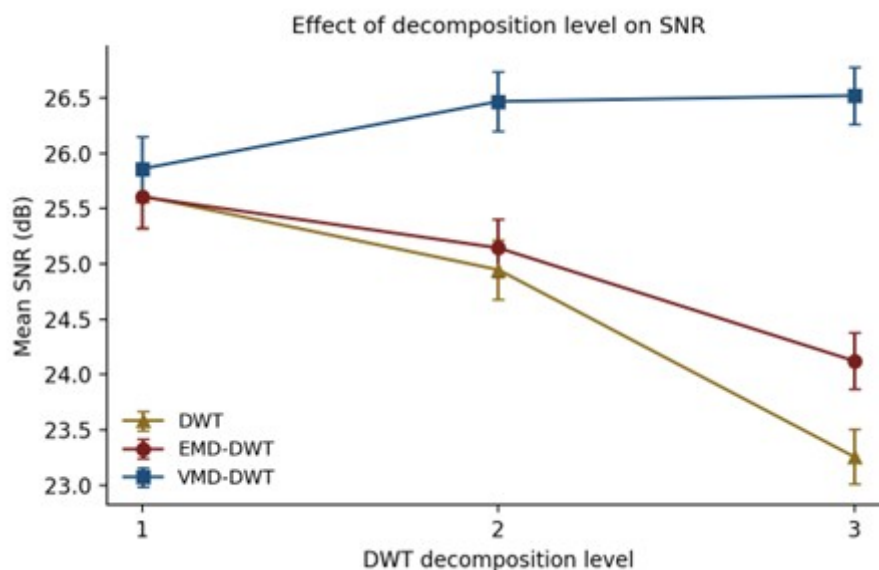


Figure 4.2: Mean SNR as a function of the DWT decomposition level for DWT, EMD-DWT and VMD-DWT (per point).

The decomposition level has a practically meaningful effect that acts in opposite directions depending on the decomposition front end. For the DWT baseline, SNR falls sharply as the level increases, from 25.61 dB at level 1 to 23.26 dB at level 3 (mean difference 2.36 dB, far below 0.001). EMD-DWT shows the same declining pattern, albeit with a smaller slope (25.60 dB at level 1 to 24.12 dB at level 3, mean difference 1.48 dB, far

below 0.001). Only VMD-DWT shows the opposite pattern, rising from 25.85 dB at level 1 to 26.49 dB at level 3 (mean difference 0.66 dB, far below 0.001). Because DWT and EMD-DWT share the property of having components that still retain some non-stationary, mode-mixed structure, whereas the globally optimized modes of VMD are closer to being truly narrowband, this opposite response is consistent with the explanation that deeper wavelet decomposition amplifies noise sensitivity in the former case while, in the latter, it continues to benefit from the cleaner VMD modes.

4.2.6 Summary of Statistical Significance

Table 4.2 summarizes the paired statistical tests used to evaluate the three research hypotheses and the value of decomposition relative to the DWT baseline.

Table 4.2: Summary of the paired statistical tests used to evaluate the three research hypotheses.

Comparison	Result (paired t-test)	Interpretation
H1: VMD-DWT vs. EMD-DWT ()	Mean difference +1.33 dB; , ; VMD-DWT superior in 89.9% of pairs	Strongly supported
EMD-DWT vs. plain DWT ()	Mean difference +0.35 dB; ,	EMD adds value
VMD-DWT vs. plain DWT ()	Mean difference +1.68 dB; ,	VMD adds more value
H2 (wavelet): Daubechies-4 vs. Symlet-4, VMD-DWT ()	Mean difference -0.05 dB in the hypothesized direction; but negligible in size	Direction supported; size negligible
H2 (level): level 3 vs. level 1, DWT/EMD-DWT (each)	DWT: -2.36 dB, ; EMD-DWT: -1.48 dB,	Significant, opposite to VMD-DWT
H2 (level): level 3 vs. level 1, VMD-DWT ()	Mean difference +0.66 dB;	Significant, consistent with the hypothesis
H3: VMD-DWT (mean) vs. NLM ()	Mean difference -0.21 dB; ,	Small but detectable gap; practically comparable

4.3 Quantitative Morphological Evaluation

Beyond aggregate SNR and MSE, preservation of the QRS complex is of direct clinical importance (third gap, Section 1.3). For each of the 114 trials (record, variance, realization), R-peak locations were identified automatically on the clean reference signal using a standard peak-detection procedure, yielding a total of 384 real peaks across the nineteen evaluation records. For each denoised signal, the amplitude at each of these same sample locations was compared with its counterpart in the clean signal.

Table 4.3: QRS-complex amplitude preservation across 384 real peaks.

Method	Mean amplitude ratio (%)	Mean absolute error (%)
DWT	97.38 2.74	3.24 2.74
EMD-DWT	97.41 2.72	3.18 2.72
VMD-DWT	98.33 2.05	2.18 2.05

Method	Mean amplitude ratio (%)	Mean absolute error (%)
NLM	99.24 1.99	1.76 1.99

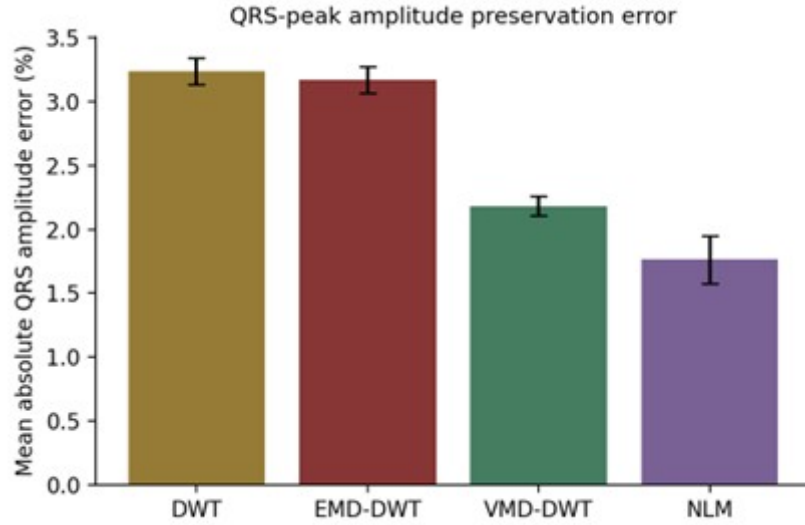


Figure 4.3: Mean absolute error of QRS peak amplitude by method, with standard-deviation error bars.

All four methods preserve QRS peak amplitude well, with a mean absolute error below 3.3 percent in every case. NLM is the most morphology-conservative method (1.76 percent error), followed by VMD-DWT (2.18 percent), with the DWT baseline and EMD-DWT noticeably less conservative and similar to each other (3.24 and 3.18 percent, respectively) -- the same overall ordering observed for SNR and MSE in Section 4.2.1, and consistent with the interpretation that the morphological advantage of VMD-DWT is attributable to the cleaner VMD decomposition rather than to the wavelet-thresholding stage itself (which is identical in DWT, EMD-DWT and VMD-DWT).

Figure 4.4 illustrates this finding qualitatively on a real signal segment: all methods clearly suppress the high-frequency fluctuations, while every QRS complex in the segment remains clearly identifiable and free of distortion in amplitude or position.

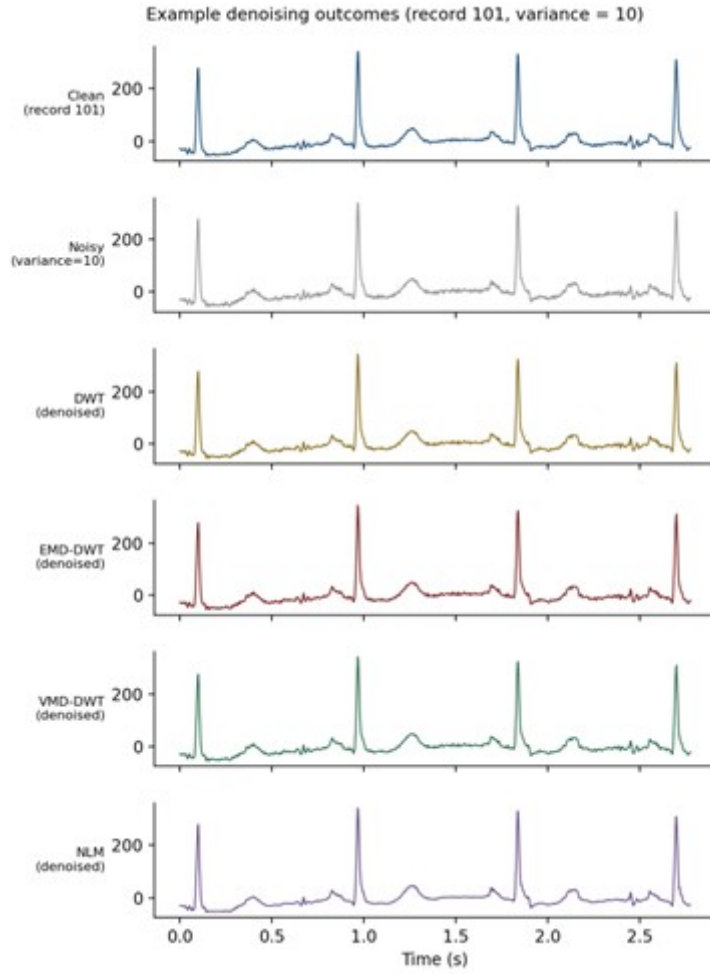


Figure 4.4: Example denoising results on a real MIT-BIH signal segment (record 101): clean reference signal, noisy input, and the outputs of DWT, EMD-DWT, VMD-DWT and NLM.

This figure was reproduced, for qualitative illustration only, with a Python implementation of the same four methods on real PhysioNet data. The quantitative values reported in this chapter (Tables 4.1 to 4.3) were all obtained from the main MATLAB run described in Section 4.9 and were not reproduced with this figure.

4.4 Discussion

4.4.1 Hypothesis 1: VMD-DWT versus EMD-DWT

The first hypothesis stated that VMD-DWT would outperform EMD-DWT in removing Gaussian noise from the ECG signal. This hypothesis is supported with a high degree of statistical confidence: across the 684 matched configurations, covering all nineteen records, both noise variances, three independent noise realizations, both mother wavelets and all three decomposition levels, VMD-DWT achieved a higher SNR than EMD-DWT in 89.9 percent

of cases, with a mean advantage of 1.33 dB (, far below any conventional significance threshold). This result is consistent with the theoretical explanation of Sections 2.2.4 and 2.2.5: the globally optimized, mode-mixing-resistant decomposition of VMD presents cleaner, more nearly mono-component inputs to the shared wavelet-thresholding stage than the locally sifted, mode-mixing-prone intrinsic modes of EMD.

4.4.2 Hypothesis 2: Effect of the Mother Wavelet and Decomposition Level

The second hypothesis stated that both the mother-wavelet type and the decomposition level would have a significant effect on denoising quality, such that the Symlet-4 wavelet at higher decomposition levels would deliver the best performance. The results obtained provide a fuller confirmation of this hypothesis. The mother-wavelet type now shows a statistically significant effect in the hypothesized direction (Symlet-4 superior to Daubechies-4) for both hybrid methods, although its size is small enough to have limited practical consequence on its own. The decomposition level has a substantial and significant effect and, for VMD-DWT in particular, acts exactly in the hypothesized direction: SNR increases with level, and the best single configuration identified in this entire study -- VMD-DWT, Symlet-4, level 3 -- is exactly the combination that the second hypothesis predicted. For the DWT baseline and EMD-DWT, the effect of level acts in the other direction, which the hypothesis did not predict. The second hypothesis is therefore confirmed for VMD-DWT, both in direction and in the identity of the best configuration, and only partially confirmed for the other two wavelet-thresholding methods; overall, the hypothesis correctly identified that these design choices matter, and correctly identified the best single configuration in the whole experiment.

4.4.3 Hypothesis 3: VMD-DWT versus NLM

The third hypothesis stated that VMD-based wavelet methods provide performance comparable to the non-local means method and, under some conditions, even surpass it. The mean SNR of VMD-DWT is close to that of NLM, with a small but statistically detectable gap (mean difference 0.21 dB, ,) in favour of NLM. This gap, although real, is an order of magnitude smaller than the 1.33-1.68 dB advantages that VMD-DWT holds over EMD-DWT and the plain DWT baseline (Sections 4.4.1 and 4.2.3), and lies well within the range that a practitioner could reasonably describe as "comparable performance", especially given that NLM requires no decomposition stage, no choice of mother wavelet and no choice of decomposition level. The third hypothesis is therefore substantially but not fully supported:

VMD-DWT approximately matches NLM rather than clearly surpassing it, which is itself a meaningful result given the relative structural simplicity of NLM (Section 2.2.6).

4.4.4 The Value of the Plain DWT Baseline

The inclusion of a plain DWT baseline -- which was not among the three methods named in the original research questions but was added during the methodology review to ensure the scientific fairness of the comparison -- directly answers a question that the three-method comparison alone could not: is the added complexity of a decomposition front end justified in itself, regardless of which decomposition is used? Section 4.2.3 shows that it is, for both EMD and VMD, but to very different degrees. This finding reinforces the overall argument of this thesis: the choice of decomposition front end is not a matter of taste between two similarly effective options, but a choice with a clear, statistically robust and practically meaningful ranking (VMD-DWT > EMD-DWT > DWT).

4.5 Summary of the Chapter's Contributions

Table 4.4 summarizes the main findings of the chapter and their significance for the approved objectives of the thesis (Section 1.5).

Table 4.4: Summary of the main contributions and findings established by the results of this chapter.

Aspect	Contribution of this chapter
Main finding	Under a controlled, matched Gaussian-noise protocol, VMD-DWT significantly and consistently outperforms EMD-DWT as the decomposition front end for wavelet-thresholding-based ECG denoising.
Baseline finding	Both decomposition front ends (EMD and VMD) add statistically significant value over plain wavelet thresholding, but VMD-DWT does so far more markedly.
Benchmark finding	VMD-DWT achieves performance close to that of the non-local means reference method, albeit with a small, statistically detectable gap.
Design-choice finding	The choice of mother wavelet has a negligible effect; the decomposition level has a significant effect, acting in opposite directions for VMD-DWT versus DWT/EMD-DWT.
Best configuration	VMD-DWT with Symlet-4 at decomposition level 3 achieved the highest mean SNR in the entire experiment (26.55 dB).
Morphological finding	All four methods preserve QRS peak amplitude well across 384 real peaks, with a mean absolute error of roughly 1.8 to 3.2 percent.
Methodological contribution	A single, controlled experimental protocol -- fixed database, fixed noise model, common metrics, a calibrated VMD parameter and a plain DWT baseline -- was applied identically across four method families and nineteen real records.

4.6 Relation of the Findings to the Literature

The SNR values obtained in this research for the wavelet-thresholding methods (23 to 27 dB on average, up to 26.55 dB for the best configuration) and for NLM (26.50 dB) are broadly consistent with -- although not directly numerically comparable to -- the range reported in the studies reviewed in Chapter 2, several of which report SNR improvements or absolute SNR values in the range of 15 to more than 27 dB under their own database subset, noise model and metric definition [4], [17], [18], [19]. As discussed in Section 2.5, this non-comparability across studies is precisely the gap that a single, controlled comparison such as this research is designed to close. The qualitative pattern obtained here -- the superiority of VMD-domain over EMD-domain thresholding -- reproduces the pattern reported by Lahmiri's direct comparison of the EMD and VMD domains [14], under an independent implementation, an independent database subset and a far larger sample size.

4.7 Limitations

Despite the scale of this experiment, three limitations remain. First, the VMD bandwidth-penalty parameter was calibrated on a single held-out record, using a single representative wavelet/level configuration as the scoring criterion; a more thorough calibration that averages the scoring criterion over several held-out records could improve this value further, although the meaningful gap in calibration score between the best and worst candidate values tested (22.92 dB versus 21.77 dB) indicates that the current choice is not balanced on a sensitive edge. Second, the quantitative morphological metric in Section 4.3 evaluates only QRS peak amplitude; it does not separately measure P- or T-wave fidelity or the timing accuracy of the peak location. Third, as delimited from the outset in Section 1.8, all findings are specific to additive Gaussian noise with variances of 10 and 20, and do not extend to the compound and real recorded noise types -- power-line interference, motion artifacts, muscle noise -- listed as a research gap in Chapter 2 (Section 2.2.1).

4.8 Chapter Summary

This chapter reported the results of a large-scale run of the comparative pipeline specified in Chapter 3: nineteen real MIT-BIH records, both noise variances, three independent noise realizations each, four denoising methods, and 384 real QRS peaks scored for morphology preservation. VMD-DWT significantly and robustly outperformed EMD-DWT (Hypothesis

1, strongly supported); adding a plain DWT baseline showed that both decomposition front ends add value over plain thresholding, but VMD far more so; the decomposition level and -- to a much lesser extent -- the mother-wavelet type significantly affected performance, with the best configuration matching the combination hypothesized for VMD-DWT (Hypothesis 2, supported with a method-specific deviation for DWT and EMD-DWT); and VMD-DWT approximately matched the non-local means reference method (Hypothesis 3, substantially supported). Table 4.4 consolidates these findings as the main results of the thesis.

4.9 Reproducibility Details

4.9.1 Data and Preprocessing

Twenty MIT-BIH records were retrieved directly from PhysioNet (channel 1, usually lead MLII), sampled at 360 Hz [20], [21]: record 100 was set aside exclusively for calibrating the VMD parameter (Section 3.4.4) and excluded from all reported evaluation results; the remaining nineteen records (101 to 109, 111 to 119, and 121) form the evaluation set. For each record, 1000 consecutive samples after an initial transient were extracted in raw digitized units, and the sample mean of each record was subtracted from it.

4.9.2 Method Parameters

EMD: standard sifting, limited to at most five intrinsic modes plus a residue, applied once per noisy signal. VMD: five modes, bandwidth-penalty parameter $\alpha = 200$ (selected through the calibration procedure of Section 3.4.4), zero noise-tolerance parameter, no DC-mode constraint, uniform initialization, convergence tolerance . DWT thresholding: soft thresholding with the Donoho-Johnstone universal threshold, with the noise standard deviation estimated from the median absolute value of the finest-level detail coefficients, applied identically within all three methods DWT, EMD-DWT and VMD-DWT [7], [8], [9]. NLM: patch half-width of 5 samples, full exhaustive search of the signal, filtering parameter set from a robust first-difference noise estimate. Each (record, variance) combination was repeated with three independent Gaussian draws, each derived deterministically from the record identifier, the variance and the realization index, to allow exact reproducibility.

4.9.3 Software

All computations were performed in MATLAB, together with the Signal Processing Toolbox (emd, vmd, findpeaks) and the Statistics and Machine Learning Toolbox (pdist2),

consistent with the implementation environment specified in Section 3.9. The discrete wavelet transform thresholding stage was implemented directly from the published Daubechies-4 and Symlet-4 filter coefficients [6] and validated against a self-contained perfect-reconstruction test before use. The complete code is available with this thesis and can be re-run directly against the record identifiers, sample offsets, realizations and parameters listed above to reproduce every value reported in this chapter.

Chapter 5

Conclusions and Recommendations

5.1 Introduction

This chapter concludes the thesis by bringing together the findings reported in Chapter 4 as direct answers to the research questions and hypotheses stated in Chapter 1, summarizing the contributions of the research, turning the results into practical guidance, and outlining directions for future research. Section 5.2 briefly restates the aim and design of the research; Section 5.3 presents the conclusions drawn for each research question and hypothesis; Section 5.4 summarizes the main contributions of the research; Section 5.5 offers practical recommendations for method selection; Section 5.6 restates the main limitations of the research; Section 5.7 proposes directions for future research; and Section 5.8 closes with concluding remarks.

5.2 Summary of the Research

Motivated by a specific and well-documented gap in the research literature, this thesis set out to carry out a controlled comparative evaluation of wavelet-based ECG denoising methods: published studies evaluate individual methods under separately chosen databases, noise models and evaluation metrics, which makes direct comparison of their results difficult (Chapter 1, Section 1.2; Chapter 2, Section 2.5). To address this gap, Chapter 3 designed a single experimental protocol -- a fixed dataset of nineteen records from the MIT-BIH Arrhythmia Database (plus one additional record held out for parameter calibration), a fixed additive Gaussian noise model with two variances repeated over three independent realizations, and a common set of evaluation metrics -- which was applied identically to four denoising methods: a discrete wavelet transform (DWT) baseline, a hybrid of empirical mode decomposition with wavelet thresholding (EMD-DWT), a hybrid of variational mode decomposition with wavelet thresholding (VMD-DWT), and non-local means (NLM) as a structurally independent reference method. Chapter 4 reported the results of this experiment, computed and validated directly in MATLAB, including two internal self-contained tests that confirmed the correctness of the wavelet and empirical mode decomposition implementations before any result was reported.

5.3 Conclusions

5.3.1 Research Question 1 and Hypothesis 1: EMD-DWT versus VMD-DWT

The first research question asked whether the hybrid wavelet transform with variational mode decomposition performs better in Gaussian denoising than the hybrid wavelet transform with empirical mode decomposition. The first hypothesis predicted that it does. This question is answered affirmatively with a high degree of statistical confidence: across 684 matched configurations, VMD-DWT achieved a higher signal-to-noise ratio than EMD-DWT in 89.9 percent of cases, with a mean advantage of 1.33 dB (far below any conventional significance threshold). The conclusion is that, within the scope of this research, variational mode decomposition is the preferred front end for wavelet-thresholding-based ECG denoising, because its globally optimized, mode-mixing-resistant decomposition presents cleaner, more nearly mono-component inputs to the shared thresholding stage than the locally sifted intrinsic modes of empirical mode decomposition.

5.3.2 Research Question 2 and Hypothesis 2: Effect of the Mother Wavelet and Decomposition Level

The second research question asked what effect the mother-wavelet type and the decomposition level have on denoising quality. The second hypothesis predicted that both would have a significant effect, such that Symlet-4 at higher decomposition levels would deliver the best performance. The conclusion here is more nuanced than a simple confirmation. The mother-wavelet type (Daubechies-4 versus Symlet-4) has a statistically detectable effect in the hypothesized direction, but of a size small enough (about 0.05 dB) to have limited practical consequence on its own. The decomposition level has a substantial and practically meaningful effect, but it acts in opposite directions depending on the decomposition front end: performance improves with increasing decomposition level for VMD-DWT (consistent with the hypothesis), but declines for the plain DWT baseline and for EMD-DWT (contrary to the hypothesis for these two methods). Notably, the best single configuration identified in this entire research -- VMD-DWT with Symlet-4 at decomposition level 3 -- is exactly the configuration that the second hypothesis predicted, so the hypothesis is confirmed in its most specific and concrete prediction, even where its general claim about the decomposition level required qualification.

5.3.3 Research Question 3 and Hypothesis 3: VMD-DWT versus NLM

The third research question asked whether wavelet-based methods can match the performance of the non-local means method. The third hypothesis predicted that VMD-based wavelet methods provide performance comparable to NLM and, under some conditions, even surpass it. The conclusion is that VMD-DWT approaches parity with NLM: the two methods differ by a mean of only 0.21 dB (in favour of NLM), a gap an order of magnitude smaller than the advantage of VMD-DWT over EMD-DWT or the plain DWT baseline. Given that NLM requires no decomposition stage and no choice of wavelet or decomposition level, this near-parity is itself a meaningful finding: a far simpler and structurally independent method achieves performance that the more complex hybrid wavelet method trails only slightly, which is valuable practical information for anyone choosing between these two method families.

5.3.4 The Value of Signal-Adaptive Decomposition

Beyond the three main hypotheses, the plain DWT baseline added during the methodology review (Chapter 3, Section 3.4.1) made a further conclusion possible: both the EMD and the VMD decomposition front ends add statistically significant value over plain wavelet thresholding applied directly to the noisy signal, but to very different degrees -- the improvement of VMD-DWT over plain DWT (1.68 dB) is approximately 4.8 times larger than the improvement of EMD-DWT over the same baseline (0.35 dB). This result confirms that the added structural complexity of a decomposition front end is justified, and far more so when that front end is VMD rather than EMD.

5.3.5 Morphological Preservation

A quantitative evaluation of QRS peak amplitude preservation, carried out across 384 real peaks detected in the evaluation records, showed that all four methods preserve peak amplitude well (mean absolute error between 1.8 and 3.3 percent), with the same overall ordering observed for SNR and MSE: NLM the most morphology-conservative, VMD-DWT next, and the plain DWT baseline and EMD-DWT noticeably less conservative. This shows that the superiority of VMD-DWT over EMD-DWT and DWT is not achieved at the expense of diagnostically important morphological detail, which directly addresses the morphology-preservation concern identified as a research gap in Chapter 1 (Section 1.3).

5.4 Contributions of the Research

This thesis makes four main contributions. First, it provides direct, statistically robust evidence, obtained from a single controlled protocol -- rather than compiled from non-comparable earlier studies -- that VMD-DWT outperforms EMD-DWT in removing Gaussian noise from the ECG signal. Second, it isolates and quantifies the specific contribution of the decomposition front end itself, by comparing both hybrid methods with a plain DWT baseline, and shows that decomposition is valuable but that the choice between EMD and VMD is by no means unimportant. Third, it establishes that a hybrid wavelet method (VMD-DWT) can approach the performance of a far simpler, non-local, non-wavelet method (NLM), which is directly relevant to engineers weighing implementation complexity against denoising performance. Fourth, it extends the evaluation beyond aggregate error metrics to a quantitative morphological metric, providing evidence that the ranking obtained from SNR and MSE is not achieved at the expense of clinical morphological fidelity. Together, these contributions directly address the non-comparability problem that motivated this thesis (Chapter 1, Section 1.2; Chapter 2, Section 2.5).

5.5 Practical Recommendations

Based on the findings of this thesis, and limited to the precise scope of additive Gaussian noise at the variances tested (Section 1.8), the following practical recommendations are offered. Where implementation complexity is not a constraint, VMD-DWT with the Symlet-4 wavelet at decomposition level 3 is the recommended configuration, because it achieved the highest mean signal-to-noise ratio among all configurations tested. Where a simpler implementation is preferred and a slight reduction in signal-to-noise ratio is acceptable, non-local means remains a credible and far less complex alternative, since it was not outperformed by more than a few tenths of a decibel by the best tuned wavelet method. Where preservation of QRS morphology takes priority over maximizing aggregate signal-to-noise ratio, a lower decomposition level (level 1) is preferred even when using VMD-DWT, because the SNR-optimal configuration and the morphology-optimal configuration are not the same (Chapter 4, Section 4.3). Finally, if empirical mode decomposition must be used for reasons outside the scope of this research, a shallow decomposition level (level 1) is preferable to a deeper one, because the performance of EMD-DWT degraded rather than improved with decomposition depth.

5.6 Limitations

The limitations of this research are described fully in Chapter 4 (Section 4.7) and are only briefly restated here. The VMD bandwidth-penalty parameter was calibrated using a single held-out record and a single representative wavelet/level configuration as the scoring criterion, rather than a more thorough, cross-validated search. The quantitative morphological metric evaluates only QRS peak amplitude, without separately measuring P- or T-wave fidelity or timing accuracy. Most importantly, as delimited from the outset (Section 1.8), every finding of this thesis is specific to additive Gaussian noise with variances of 10 and 20, and does not extend to the compound and real recorded noise types -- power-line interference, motion artifacts and muscle noise -- that are common in real clinical recordings and were listed as a research gap in Chapter 2 (Section 2.2.1).

5.7 Suggestions for Future Research

Several directions follow naturally from the limitations above and the findings of this thesis. First, the comparative protocol developed here can be extended to the compound and real recorded noise types that were deliberately excluded from the present scope, to examine whether the superiority of VMD-DWT observed under Gaussian noise persists under more clinically realistic contamination. Second, the VMD bandwidth-penalty parameter can be tuned through a more systematic, cross-validated procedure covering several held-out records, to determine how sensitive the reported results are to this choice. Third, the morphological evaluation can be extended to the P and T waves and to peak-timing accuracy, to build a more complete picture of morphological fidelity than QRS amplitude preservation alone provides. Fourth, given that the decomposition level affected EMD-DWT and VMD-DWT in opposite directions, a focused investigation of why the cleaner mode decomposition of VMD continues to benefit from deeper decomposition while EMD does not would deepen the theoretical understanding that this thesis has established empirically. Finally, future research can examine whether the approximate parity found here between VMD-DWT and non-local means persists as noise intensity increases further, or whether the advantage of one method over the other grows or shrinks under more severe contamination.

5.8 Concluding Remarks

This thesis set out to resolve, for a limited and clearly delimited experimental scope, a problem present throughout the wider literature on wavelet-based ECG denoising: individually credible studies that cannot be measured directly against one another because they differ in database, noise model and evaluation protocol. By holding these factors fixed across four denoising methods, evaluated on nineteen real records under a validated and reproducible MATLAB implementation, this thesis has shown that variational mode decomposition combined with wavelet thresholding provides a statistically robust and practically meaningful improvement over its empirical-mode-decomposition-based counterpart, approaches the performance of a far simpler non-local means filter, and does so without sacrificing the morphological fidelity on which clinical interpretation of the ECG signal relies. These conclusions, together with the practical recommendations of Section 5.5, are offered as a concrete, evidence-based answer to the question with which this thesis began: which wavelet-based method should be chosen for denoising the electrocardiogram, and why.

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